Generating Functions 1

Siddharth K

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Problem 1

If $|\alpha|_{\mathcal{B}} = 2|\alpha|_{\mathcal{A}}$, the coefficients are the same but the exponents are doubled, so $B(z) = A(z^2)$.

Problem 2

$$\mathrm{SEQ}_{\mathrm{even}}(z) = 1 + B(z)^2 + B(z)^4 + \cdots = \frac{1}{1 - B(z)^2},$$

$$\mathrm{SEQ}_{\mathrm{odd}}(z) = B(z) + B(z)^3 + B(z)^5 + \cdots = \frac{B(z)}{1 - B(z)^2}.$$

Problem 3

We need to form a bijection between multisets of $\mathcal{C}$ of size n and multisets of $\mathcal{C}$ of size $2n$ along with a subset of $\mathcal{C}$. To do this, take any multiset of $\mathcal{C}$, and consider the multiplicity of each element. If it is odd, put the element into P and subtract that 1, so the remaining multiplicity is even. Then halve the remaining multiplicity and use that for M. The first action contributes $P(z)$ and the second contributes $M(z^2)$, since each copy in the second multiset represents a pair of copies of the element c, so it contributes $z^{2\cdot|c|}$. This is reversible, since from the subset and multiset, we can recover the original multiplicity by multiplying by 2 and adding 1 if the element is in the subset. Therefore, $M(z) = P(z)M(z^2)$.

If we repeatedly iterate this, we get

$$M(z) = P(z)M(z^2),$$

$$M(z^2) = P(z^2)M(z^4) \implies M(z) = P(z)P(z^2)M(z^4)$$

$$= P(z)P(z^2)P(z^4)M(z^8) = \cdots$$

$$= \prod_{n=0}^{\infty} P(z^{2^n}).$$

Problem 4

Consider sequences of length n. There are n different rotations we could perform on any sequence that correspond to the same cycle. By Burnside's lemma, the number of unique cycles is

$$\frac{1}{n}\sum_{j=0}^{n-1}|\text{Fix(rotate by } j)|.$$

The generating function representing sequences of length n that are fixed under rotation by j is

$$B(z^{n/\gcd(n,j)})^{\gcd(n,j)},$$

since $\gcd(n,j)$ objects from $\mathcal{B}$ can be chosen to repeat $n/\gcd(n,j)$ times each. Plugging into Burnside's lemma gives us

$$\frac{1}{n}\sum_{j=0}^{n-1}B(z^{n/\gcd(n,j)})^{\gcd(n,j)}.$$

We can reindex this by summing over the divisors of n:

$$\frac{1}{n}\sum_{d|n}\phi(n/d)B(z^{n/d})^d,$$

since for each d, the number of j such that $\gcd(n,j)=d$ is the same as the amount of numbers $< n/j$ relatively prime to n/j, which is $\phi(n/d)$. Finally, we sum over n and simplify:

$$\begin{aligned}
\text{CYC}(\mathcal{B})(z) &= \sum_{n\geq 1}\frac{1}{n}\sum_{d|n}\phi(n/d)B(z^{n/d})^d \\
&= \sum_{n\geq 1}\sum_{d|n}\frac{1}{n}\phi(d)B(z^d)^{n/d} \\
&= \sum_{d\geq 1}\sum_{m\geq 1}\frac{1}{dm}\phi(d)B(z^d)^m \\
&= \sum_{d\geq 1}\frac{\phi(d)}{d}\sum_{m\geq 1}\frac{B(z^d)^m}{m} \\
&= \sum_{d\geq 1}\frac{\phi(d)}{d}\log\frac{1}{1-B(z^d)}.
\end{aligned}$$

Problem 6

Since n is always at the peak, for each number from $n-1$ to 1 in order, we can choose either left or right to produce a unique unimodal permutation. So the

count is 2^{n-1}, and the generating function is

$$\sum_{n\geq 1} 2^{n-1} z^n = \frac{z}{1-2z}.$$

Problem 7

The generating function is

$$C(z) = \frac{1}{1-z-z^2}.$$

Taking the partial fraction decomposition and and unsumming each term, we get that the n'th coefficient is

$$C_n = \frac{1}{\sqrt{5}}(\varphi^{n+1} - \overline{\varphi}^{n+1}),$$

which is the $n+1$'th Fibonacci number. This is because $\mathrm{SEQ}(\mathcal{Z} + \mathcal{Z}^2)$ can be represented by making a sequence of 1s and 2s, so an object of total size n is a way of writing n as an ordered sum of 1s and 2s, which satisfies the Fibonacci recurrence.

Problem 9

$$\frac{A(z)}{1-z} = A(z)\sum_{n=0}^{\infty} z^n = \left(\sum_{n\geq 0} a_n z^n\right)\left(\sum_{n\geq 0} z^n\right) = \sum_{m\geq 0}\sum_{n\geq 0} a_m z^{m+n}$$
$$= \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} a_k\right) z^n,$$

so the coefficients are $\sum_{k=0}^{n} a_k$. In terms of admissible constructions, we can think of $A(z)\cdot\frac{1}{1-z}$ as the Cartesian product $\mathcal{A}\times\mathrm{SEQ}(\mathcal{Z})$. If we choose an object from $\alpha \in \mathcal{A}$ with size k, there is exactly one object from $\mathrm{SEQ}(\mathcal{Z})$ that can be paired with α so that the total size is n. So the number of objects of size n in $\mathcal{A}\times\mathrm{SEQ}(\mathcal{Z})$ is $\sum_{k=0}^{n} a_k$.

Problem 10

Given a valid string of length $n-1$, we can get a valid string of length n by appending a $0, 1$, or 2, but only a 2 when the string doesn't end in 0. If a_n is the number of valid strings of length n, then a_{n-1} of them end in 0, since you can take any valid $n-1$ length string and append 0. So we get the recurrence

$$a_{n+1} = 3a_n - a_{n-1}.$$

To turn this into a generating function, we multiply by z^{n+1} and sum over n:

$$\begin{aligned}
\sum_{n\geq 1} a_{n+1} z^{n+1} &= 3\sum_{n\geq 1} a_n z^{n+1} - \sum_{n\geq 1} a_{n-1} z^{n+1}, \\
\sum_{n\geq 1} a_{n+1} z^{n+1} &= A(z) - a_0 - a_1 z = A(z) - 1 - 3z, \\
3\sum_{n\geq 1} a_n z^{n+1} &= 3z(A(z) - 1), \\
\sum_{n\geq 1} a_{n-1} z^{n+1} &= z^2 A(z), \\
A(z) - 1 - 3z &= 3z(A(z) - 1) - z^2 A(z), \\
A(z)(1 - 3z + z^2) &= 1, \\
A(z) &= \frac{1}{1 - 3z + z^2}.
\end{aligned}$$